ESG ANALYTICS · SEBI BRSR

Carbon Intensity of the Nifty 50

An audit of India's mandatory sustainability filings for 49 of the largest listed companies, and a stress test of a regression-based carbon grading model on real data.

1. Data and methodology
2. Data quality audit
3. Findings
4. Evaluating the (m)y grading model
5. Recommendations
6. Limitations and next steps

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DATA
SEBI BRSR via NSE, FY23-FY24

TOOLS
Python · pandas · scipy · SQL (DuckDB: CTEs, window functions)

Analyst	Mohammad Faiz Ahmad	**Status**	Final
Data source	SEBI BRSR, Principle 6 (XBRL + PDF), via NSE	**Period**	FY2022-23 and FY2023-24
Coverage	49 of 50 Nifty 50 companies, 98 company-years	**Version**	v1.0, October 2026

Executive summary

Questions. (1) Using only what companies are legally required to disclose, who emits the most carbon per rupee earned, and is it getting better? (2) Does a regression-based "carbon grade" tell us anything that a simple ratio doesn't?

Answer in one line. The raw filings can't be trusted as filed. Once they're fixed, carbon in the Nifty 50 is a story about five companies and one sector divide. The improvement in the headline numbers comes from revenue growth, and the grading model is too unstable to use.

39%
of machine-readable filings had unit or consistency errors

52%
of all Scope 1+2 emissions come from NTPC alone (86% from the top 5)

–9.3% / +2.8%
median intensity vs median absolute emissions, year on year

32 / 49
model grades flip under one alternative scaling choice

Key findings

1. **The data needs an audit before any analysis.** 19 of 49 filings had errors. NTPC's 352 Mt was tagged as tonnes, which would have made India's biggest emitter look like its cleanest.
2. **Sector explains 74% of the variance in intensity.** Power has a median of 977 tCO2e per Rs crore; financial services has 0.74. Cross-industry rankings mostly rank industries.
3. **Emissions are extremely concentrated.** Five companies emit 85.8% of the 675 Mt total, and NTPC alone emits 52%.
4. **The "improvement" is revenue growth.** Intensity fell for 34 of 44 companies, but absolute emissions rose.
5. **The (m)y model only partly agrees with intensity** (Spearman ρ = –0.55) and is unstable, so I would not ship it as a rating.

1 · Data and methodology

Collect
Filing index from NSE's API; downloaded each BRSR XBRL file and PDF

→

Extract
Parsed Scope 1, Scope 2, energy and revenue from Principle 6

→

Audit
Compared every unit with the company's own PDF and logged each fix

→

Analyse
Intensity, YoY change, sector z-scores, concentration

→

Evaluate
(m)y grading model plus 6 sensitivity variants

Data dictionary

Field	Meaning
`scope1_tco2e`	Direct emissions: fuel the company burns itself
`scope2_tco2e`	Indirect emissions from purchased electricity
`energy_gj`	Total energy consumed
`energy_renewable_gj`	Energy from renewable sources
`revenue_cr`	Revenue from operations (Rs crore), from the same filing
`source_url`	Link to the NSE file each value came from

Metrics

Metric	Definition
Intensity	(Scope 1 + Scope 2) ÷ revenue, tCO2e per Rs crore
Scope 1 share	Scope 1 ÷ (Scope 1 + Scope 2)
YoY change	FY24 vs FY23, excluding breaks above 3x
Sector z-score	z-score of log10 intensity within a sector (sectors with 3+ companies only)

Emissions and revenue always come from the same filing, so they share the same boundary (33 standalone, 16 consolidated). Mahindra & Mahindra is excluded because its files return 404 on NSE.

2 · Data quality audit

Before any analysis, I checked every machine-readable value against the human-readable PDF. **19 of 49 filings (39%) had at least one value that couldn't be used as tagged.**

Error type	Companies	Who	Example / impact
Wrong emissions unit	3	NTPC, Tata Steel, ITC	NTPC filed 352 *million* tonnes under a "tonnes" tag, understating it by 1,000,000x
Wrong energy unit	8	NTPC, ITC, UltraTech, Adani Ports, TCS, Bajaj Auto, BSE, Axis Bank	TJ, MJ or PJ filed under a GJ tag; Axis's XBRL value doesn't match its own PDF
Wrong revenue unit	8	Incl. HCL (Rs million)	Rs crore typed where rupees were expected
Methodology breaks	4	Apollo, Tata Consumer, SBI Life, Axis	Apollo's Scope 1 went from 8,033 t to 731,919 t; excluded from YoY only

> **Rule I followed:** no value is estimated. Every number is either as filed or converted using the unit stated in the company's own PDF, and each conversion is logged with a link in `data/corrections.csv`. Hindalco (whose PDF refers to its annual report), Coal India and Cipla (both with unusually high emissions per unit of energy) are kept as filed and flagged.

3 · Findings

3.1 Industry decides almost everything

Intensity spans **144,820x**, from Eternal at 0.013 to NTPC at 1,946 tCO2e per Rs crore; the median company is at 5.2. **Sector alone explains 74%** of the variance in log intensity.

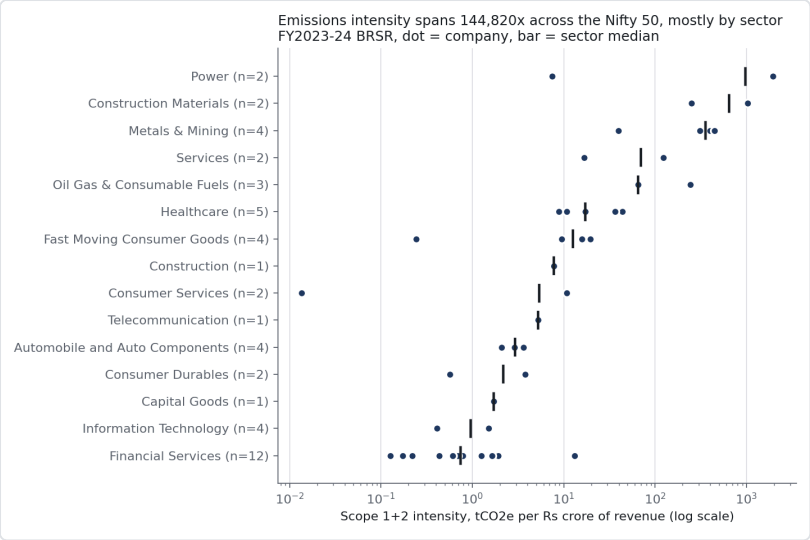


Figure 1. Scope 1+2 intensity by sector (log scale). Dot = company, bar = sector median.

Sector	Median
Power	976.8
Construction Materials	641.2
Metals & Mining	353.9
Oil, Gas & Fuels	65.5
Healthcare	17.2
FMCG	12.6
Automobile	2.9
IT	0.96
Financial Services	0.74

tCO2e per Rs crore, FY24, selected sectors

3.2 Five companies are the whole story

#	Company	Sector	Scope 1+2 (Mt)	Share of total	Intensity	(m)y grade
1	**NTPC**	**Power**	**352.5**	**52.2%**	**1,946**	**Brown**
2	UltraTech Cement	Construction Materials	73.1	10.8%	1,031	Brown
3	Tata Steel	Metals & Mining	63.0	9.3%	447	Neutral
4	JSW Steel	Metals & Mining	53.2	7.9%	398	Brown
5	Reliance	Oil, Gas & Fuels	37.7	5.6%	66	Neutral
6	Coal India	Oil, Gas & Fuels	34.7	5.1%	244	Neutral
7	Hindalco	Metals & Mining	25.7	3.8%	310	Brown
Total, 49 companies			**675.3**	**100%**	The 12 financial firms: 0.28%	

→ Any decarbonisation conversation about the Nifty 50 is really about five to seven companies. Banks look clean only because financed (Scope 3) emissions aren't required to be reported.

> **3.3 The "improvement" is mostly revenue growth.** Across 44 clean year-on-year pairs, median intensity **fell 9.3%** (34 of 44 companies improved), but median absolute emissions **rose 2.5%** (45 pairs without an emissions break; +2.8% on the same 44). Intensity is the number companies prefer to report. The atmosphere sees the absolute number.

3.4 Who is dirty *for their sector*?

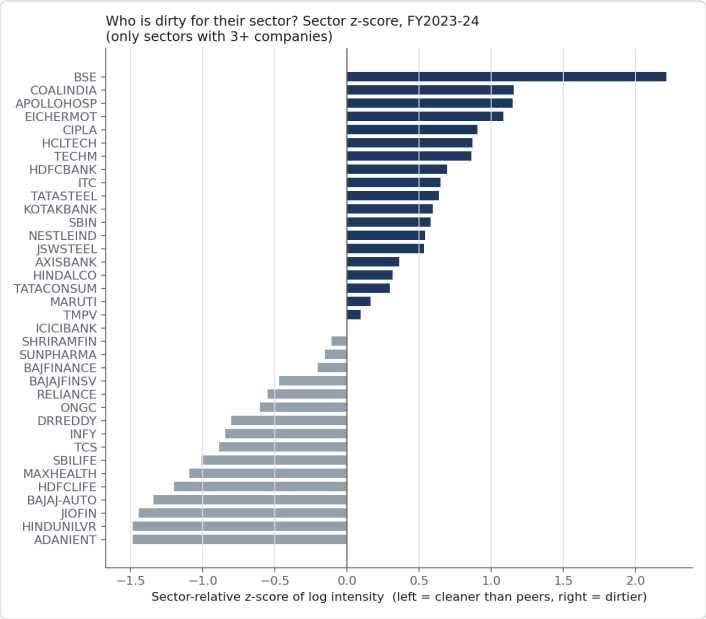


Figure 2. Each company vs its own sector (z-score of log intensity). Right = dirtier than peers.

A fair comparison is within a sector, so I z-scored each company's log intensity inside its industry (only industries with 3+ companies).

- **Dirtier than peers:** BSE (+2.2; its purchased power is large relative to its revenue), Coal India (+1.15), Apollo Hospitals (+1.15), Eicher (+1.08)
- **Cleaner than peers:** Adani Enterprises (–1.49), HUL (–1.48, almost entirely renewable power), Bajaj Auto (–1.34)

→ **This is the view I'd put in front of a decision-maker, not the raw ranking.**

4 · Evaluating the (m)y grading model

The (m)y model was proposed by a senior as a concept, without data. Score five facts about a company from 1 to 10, fit a straight line through them in a fixed order, and grade the company by the slope **m** and intercept **y**: **Acceptable** the line slopes up · **Neutral** flat · **Brown exposure** slopes down.

Point	Deck factor	BRSR field I mapped it to	Scaling
1	Input	Total energy consumed	log10, min-max to 1-10
2	Input	Non-renewable share of energy	min-max to 1-10
3 · 4	Transition	Scope 1 · Scope 2 emissions	log10, min-max to 1-10
5	Demand	Revenue from operations	log10, min-max to 1-10

The deck gives no numeric thresholds, so I set the bands myself: Acceptable if m > 0.25, Neutral if |m| ≤ 0.25, Brown if m < –0.25.

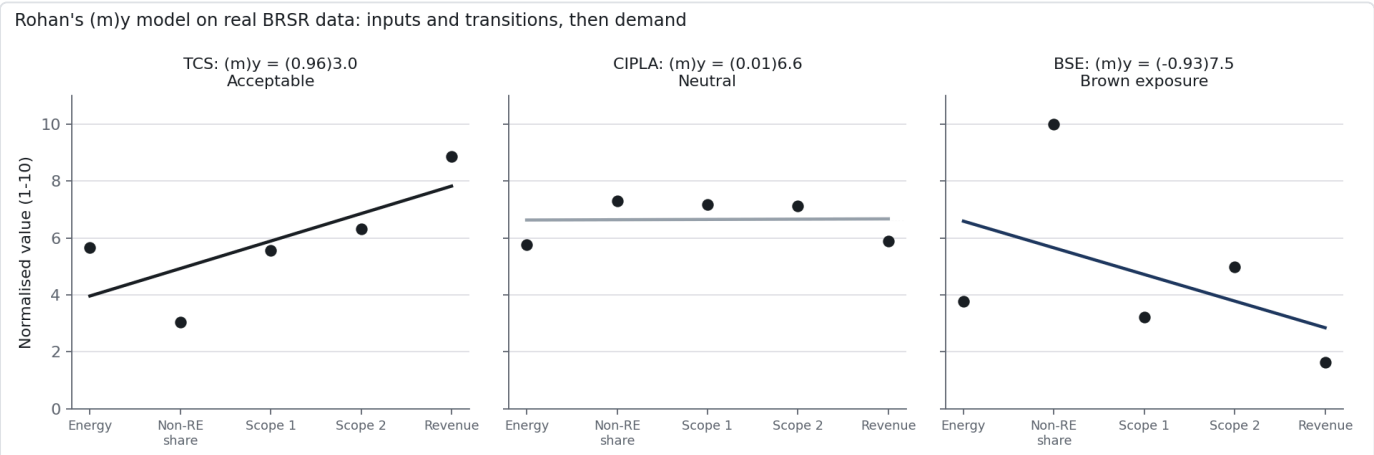


Figure 3. The model on real data for three companies: TCS slopes up (Acceptable), Cipla is flat (Neutral), BSE slopes down (Brown exposure).

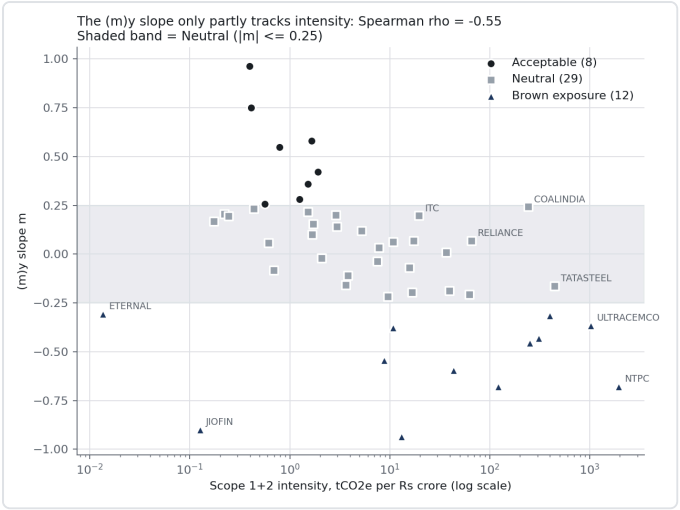


Figure 4. If the model worked, high-intensity companies (right) would all be Brown (bottom). Coal India, Reliance and Tata Steel sit in Neutral; Eternal and Jio Financial, two of the three cleanest, are Brown.

Verdict: it only partly works

- Grades: 8 Acceptable, 29 Neutral, 12 Brown exposure
- Agrees with intensity only moderately: **Spearman ρ = –0.55** ($p < 0.001$)
- Coal India ranks **43rd of 49** on intensity but is **9th best** by slope

Why it breaks

- **Size cancels out.** Big firms have big emissions *and* big revenue, so the line goes flat. The slope partly measures size.
- **Relative scaling.** A company's grade depends on who else is in the set.
- **Scope 3 is missing**, so banks look clean.
- **Sector mix** dominates every cross-industry comparison.

Sensitivity: how stable are the grades?

Variant	Acceptable	Neutral	Brown	Grades changed	ρ vs intensity
Default (log min-max, band 0.25)	**8**	**29**	**12**	**0**	**–0.55**
Linear min-max scaling	4	6	39	32	–0.07
Percentile-rank scaling	15	18	16	15	–0.47
Neutral band 0.10	20	10	19	19	–0.55
Neutral band 0.50	4	39	6	10	–0.55
Drop non-renewable share	25	15	9	21	–0.77
Drop energy	6	10	33	26	–0.21

→ A rating that changes for 32 of 49 companies when one modelling choice changes is not a rating. If anything, dropping the non-renewable-share point makes it track intensity better (ρ = –0.77).

5 · Recommendations

For…	Recommendation
Anyone using BRSR data	Never trust the XBRL as filed. Run a unit audit against the PDF first, because 39% of filings fail it.
Ranking companies	Use sector-relative intensity, not one cross-industry list.
Tracking progress	Report absolute emissions next to intensity. Intensity alone hides a 2.8% rise in absolute emissions.
Composite "grades"	Test a model's stability before shipping it. This one fails on size bias and scaling choices; use intensity plus the sector z-score instead.

6 · Limitations and next steps

Limitations

- Scope 3 is optional in BRSR, so financed and product-use emissions are missing
- Only two years, so trends are a single change, not a series
- With 49 companies, 8 of 15 sectors have fewer than 3 companies, which is too few for peer z-scores
- Standalone and consolidated boundaries are mixed across companies

Next steps

- Extend to the top 1,000 BRSR filers for proper sector peer groups
- Add Scope 3 where it is disclosed
- Re-run on FY25 filings to test whether the decoupling holds
- Turn the unit audit into an automated check that flags suspicious filings

Appendix · Reproducibility

Step	Command / file
Extract and correct	`python extract.py` → `data/brsr_emissions.csv` (applies `data/corrections.csv`)
PDF unit audit	`python audit_units.py` → `data/pdf_unit_audit.csv`
Analysis and charts	`python analysis.py` → `data/company_metrics.csv`, `charts/`
SQL cross-check	`python run_sql.py` → `sql/results/` (DuckDB CTEs and window functions; asserts the results match the Python)

Builds on the (m)y model concept by Rohan Choudhary (IIT Patna), used with permission. The data pipeline, unit audit, metrics, model implementation, sensitivity tests and findings are new work. Every data row links to its NSE source file.